Integrating to Infinity

Up to now, we've only dealt with integrals over a *finite* domain $[a, b]$. In fact, if you look back to when we proved FTC I, you'll see that it only deals with finite domains...

Definition Integrating over infinite domains (Part 1):
If $f(x)$ is continuous on $(-\infty, \infty)$ then

$$\int_a^\infty f(x)\,dx = \lim_{b\to\infty}\int_a^b f(x)\,dx \qquad \text{and} \qquad \int_{-\infty}^b f(x)\,dx = \lim_{a\to-\infty}\int_a^b f(x)\,dx.$$

These integrals are said to *converge* if the limits exist and are finite.

Examples

Compute the following integrals:
$\int_0^\infty e^{-2x}\,dx$

$\int_1^\infty \frac{1}{x^2}\,dx$

$\int_{-\infty}^{-1} \frac{1}{x}\,dx$

Definition Integrating over infinite domains (Part 2):

If $f(x)$ is continuous on $(-\infty, \infty)$ then

$$\int_{-\infty}^{\infty} f(x)\, dx = \int_{-\infty}^{a} f(x)\, dx + \int_{a}^{\infty} f(x)\, dx,$$

where a is any number. This integral only converges (i.e. only exists) when both integrals in the sum exist and are finite.

Example

Compute the integral $\displaystyle\int_{-\infty}^{\infty} \frac{1}{1+x^2}\, dx$.

Comparing Integrals

Sometimes we can't find an antiderivative to solve an integral completely, but we can make the integral a little simpler and check whether the original diverges or converges.

Examples

$\int_2^\infty \frac{1}{\sqrt{x^2-1}}\,dx$

(Hint: $\sqrt{x^2-1}$ is a little smaller than $\sqrt{x^2} = x$. What does that tell you about $\frac{1}{\sqrt{x^2-1}}$ compared to $\frac{1}{x}$?)

Does $\int_2^\infty \frac{1}{\ln x}\,dx$ converge or diverge?

(Hint: consider the linear approximation to $\ln x$ at $x = 1$. How does it compare to $\ln x$ itself?)

Exercises

Calculate the following integrals, if they converge.

1. $\int_0^\infty \frac{e^x}{(e^x+1)^2}\,dx$ (This is a substitution. Don't forget to change bounds!)

□Regrade

2. $\int_0^\infty xe^{-x}\,dx$ (You may need L'Hopital's rule somewhere along the way...)

□Regrade

3. $\int_0^{\infty} \frac{1}{(x+4)^2}\, dx$

$\square_{\text{Regrade}}$

4. $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \csc^2(x)\, dx$ (Hint: $\csc^2(x) = \frac{\sec^2(x)}{\tan^2(x)}$)

$\square_{\text{Regrade}}$

5. The mass of pollutants over city up to height u meters is given by $\int_0^u 25,600\pi e^{-0.0025h}\,dh$ kilograms. Compute the total mass of pollutants over the city. (Compare Varying Density Lab, Question 15.)

□Regrade

6. Find c such that $\int_{-\infty}^{\infty} f(t)\,dt = 1$

□Regrade

$$f(t) = \begin{cases} cte^{-\frac{t}{2}} & t > 0 \\ 0 & \text{otherwise} \end{cases}$$

For the next two questions, don't attempt to compute the integral. Instead, compare each of them to other integrals to see if they converge or diverge.

7. $\int_0^\infty \frac{1}{e^x + 2^x}\, dx$

□Regrade

8. $\int_1^\infty \frac{1 + \sin x}{x^2}\, dx$

□Regrade

(Hint: _____ $\leq \sin(x) \leq$ ____. How does that help you bound the integrand?)

9. Draw the graph of x^3. What do you think $\int_{-\infty}^{\infty} x^3\,dx$ *should* be? Show that it in fact $\square^{\text{Regrade}}$ diverges instead.